

AN EXACT TRANSFER-MATRIX PROOF OF BARKER'S ORDER-TEN RECURRENCE FOR OEIS A321614

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ABSTRACT. OEIS A321614 counts maximum nonattacking king placements on a $4 \times 2n$ board modulo the four generic rectangle symmetries. Colin Barker conjectured a rational generating function and an order-ten recurrence from the first 22 terms. We prove the conjecture for every n . Equality in the elementary 2×2 packing bound reduces the placements to paths in a twelve-state graph. An explicit polynomial matrix identity over the integers determines all four Burnside fixed-point generating functions. Their average is Barker's function, and an exact coprimality certificate proves that its degree-ten denominator is minimal. We also isolate a necessary convention: at $n = 2$ the board is square, but A321614 retains the order-four rectangle subgroup; the full square group would give 14 rather than 23 orbits. The proof is accompanied by two independent exact verifiers.

1. INTRODUCTION

The OEIS entry A321614 defines $a(n)$ as the number of nonequivalent ways to place the maximum number $2n$ of nonattacking kings on a $4 \times 2n$ board [3]. Its formula section records a rational generating function and a tenth-order recurrence as conjectures contributed by Colin Barker in 2018. The published sequence data stop at $n = 21$.

Maximum king placements on fixed-width even boards have a classical finite state structure. Wilf used the forced occupancy of 2×2 cells to study the labeled counts and proved a general fixed-height asymptotic theorem [4]. In the present height-four case, the labeled identity branch alone is OEIS A061593 [2]. The extra issue for A321614 is to determine, without losing odd-length or center-seam cases, the three nonidentity Burnside branches.

A 2026 exact computation reproduced the 22 published terms and verified Barker's recurrence through $n = 5000$, while explicitly leaving an all- n transfer proof open [1]. The argument below supplies that proof. Its decisive step is the finite integer identity (7), not an inference from any computed prefix.

2. DEFINITION AND THE TWELVE STATES

For $n \geq 1$, let

$$B_n = \{0, 1, 2, 3\} \times \{1, \dots, 2n\}.$$

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Two distinct squares attack when their Chebyshev distance is at most one. Let X_n be the set of nonattacking subsets of B_n having $2n$ elements. We use the four-element group

$$G = \{1, h, v, hv\}, \quad h(r, c) = (3 - r, c), \quad v(r, c) = (r, 2n + 1 - c),$$

for every n , including $n = 2$. Define $a(n) = |X_n/G|$ and set $a(0) = 1$.

Partition the board into the $2n$ disjoint 2×2 cells obtained from the two row pairs and the n consecutive column pairs. Each cell contains at most one king, so $2n$ is an upper bound. It is attained, for example, by

$$\{(0, 2j - 1), (2, 2j - 1) : 1 \leq j \leq n\}.$$

Equality forces every one of those 2×2 cells to contain a king.

Encode a column by its four-bit occupancy mask. The legal masks are

$$\mathcal{C} = \{0, 1, 2, 4, 5, 8, 9, 10\}.$$

For $p, q \in \mathcal{C}$, write $p \sim q$ if the occupied rows in two adjacent columns do not attack. The two-column blocks of total weight two are exactly

$$(1) \quad \mathcal{B} = ((0, 5), (0, 9), (0, 10), (1, 4), (1, 8), (2, 8), \\ (4, 1), (5, 0), (8, 1), (8, 2), (9, 0), (10, 0)).$$

Define the 12×12 zero-one matrix T , indexed in the order (1), by

$$(2) \quad T_{(p,q),(r,s)} = [q \sim r].$$

The forced occupancy above is a bijection between X_n and length- n paths in T . With $\mathbf{1}$ the all-ones column,

$$(3) \quad I_0 = 1, \quad I_n = \mathbf{1}^T T^{n-1} \mathbf{1} \quad (n \geq 1),$$

where I_n is the identity fixed count.

3. BURNSIDE FIXED PATHS

Let ρ reverse the four row bits and define involutions on $\mathcal{B}$ by

$$h_0(p, q) = (\rho p, \rho q), \quad j_v(p, q) = (q, p), \quad j_r(p, q) = (\rho q, \rho p).$$

Top-bottom reflection acts pointwise by h_0 . Left-right reflection reverses the block word and applies j_v ; the half-turn reverses it and applies j_r . The compatibility definition immediately gives

$$(4) \quad T_{p,q} = T_{j_v q, j_v p} = T_{j_r q, j_r p}.$$

The only h_0 -fixed blocks are $(0, 9)$ and $(9, 0)$. Their induced transfer matrix is

$$T_h = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

up to interchanging the two states. Hence the horizontal fixed count and its generating function are

$$(5) \quad H_n = n + 1, \quad H(x) = \sum_{n \geq 0} H_n x^n = \frac{1}{(1-x)^2}.$$

For $j = j_v$ or j_r , define column vectors

$$d_j(p) = T_{p,jp}, \quad f_j(p) = [jp = p].$$

A j -fixed path of length $2m$ is uniquely determined by its first half and its middle edge. A fixed path of length $2m + 1$ is uniquely determined by its first half and a j -fixed center. Thus

$$(6) \quad F_{2m}^j = \mathbf{1}^T T^{m-1} d_j \ (m \geq 1), \quad F_{2m+1}^j = \mathbf{1}^T T^m f_j \ (m \geq 0).$$

Here d_v has three entries equal to one and $f_v = 0$; d_r has seven such entries and f_r has two.

4. THE EXACT RESOLVENT CERTIFICATE

Set

$$D_0(y) = 1 - 9y + 28y^2 - 33y^3 + 9y^4 = (1 - 3y)^2(1 - 3y + y^2).$$

Let $Z(y) = z_0 + z_1y + z_2y^2 + z_3y^3$, where, in the order (1),

$$\begin{aligned} z_0 &= (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1), \\ z_1 &= (3, 3, 3, -3, -3, -4, -4, -6, -3, -3, -6, -6), \\ z_2 &= (-1, -1, -1, 0, 0, 3, 3, 10, 0, 0, 10, 10), \\ z_3 &= (0, 0, 0, 0, 0, 0, -3, 0, 0, -3, -3). \end{aligned}$$

The central certificate is the coefficientwise integer matrix identity

$$(7) \quad \boxed{Z(y)(I - yT) = D_0(y)\mathbf{1}^T}.$$

Because $D_0(0) = 1$, it is an identity of formal power series and gives

$$\mathbf{1}^T(I - yT)^{-1} = \frac{Z(y)}{D_0(y)}.$$

The sums of the four displayed rows are 12, -29, 33, -9. Equations (1), (7), and (3) therefore yield

$$(8) \quad I(x) = \sum_{n \geq 0} I_n x^n = \frac{1 + 3x - x^2}{(1 - 3x)^2(1 - 3x + x^2)}.$$

Direct scalar products in the same integer certificate are

$$\begin{aligned} Z(y)d_v &= 3 - 18y + 30y^2 - 9y^3, \\ Z(y)d_r &= 7 - 32y + 36y^2 - 9y^3, \quad Z(y)f_r = 2 - 6y. \end{aligned}$$

Substituting $y = x^2$ in (6) and cancelling exact factors gives

$$(9) \quad V(x) = \sum_{n \geq 0} V_n x^n = \frac{1}{1 - 3x^2},$$

and

$$(10) \quad R(x) = \sum_{n \geq 0} R_n x^n = \frac{1 + x + x^2}{(1 - x - x^2)(1 - 3x^2)}.$$

Thus all four branches have been established for every length by finite integer equalities.

5. THE RECURRENCE AND ITS MINIMALITY

Theorem 5.1. *The ordinary generating function of the orbit counts defined in Section 2 is*

$$(11) \quad \sum_{n \geq 0} a(n)x^n = \frac{(1-2x)(1-6x+17x^2-18x^3-2x^4+7x^5+6x^6-3x^7)}{(1-x)^2(1-3x)^2(1-3x+x^2)(1-x-x^2)(1-3x^2)}.$$

This fraction is reduced over $\mathbb{Q}(x)$.

Proof. Burnside's lemma, including the formal empty case, gives

$$\sum_{n \geq 0} a(n)x^n = \frac{I(x) + H(x) + V(x) + R(x)}{4}.$$

Substitution from (5), (8), (9), and (10), followed by polynomial addition, gives (11).

Let its numerator and denominator be N and D . The irreducible factors of D over $\mathbb{Q}$ are $1-x$, $1-3x$, and the three displayed quadratics (with discriminants 5, 5, 12). The remainders of N modulo these five factors are, respectively,

$$-2, \quad \frac{170}{2187}, \quad -37 + 97x, \quad 275 - 445x, \quad \frac{128}{9} - \frac{74}{3}x.$$

None vanishes, so $\gcd(N, D) = 1$. □

Expanding the denominator gives

$$\begin{aligned} D(x) = & 1 - 12x + 54x^2 - 98x^3 - 17x^4 + 346x^5 - 505x^6 \\ & + 210x^7 + 120x^8 - 126x^9 + 27x^{10}. \end{aligned}$$

Since $\deg N = 8$, coefficient comparison proves, for every $n \geq 10$,

$$\begin{aligned} a(n) = & 12a(n-1) - 54a(n-2) + 98a(n-3) + 17a(n-4) \\ & - 346a(n-5) + 505a(n-6) - 210a(n-7) - 120a(n-8) \\ & + 126a(n-9) - 27a(n-10). \end{aligned}$$

The reduced denominator has degree ten, so no eventual homogeneous constant-coefficient recurrence over $\mathbb{Q}$ can have smaller order.

6. THE SQUARE BOUNDARY

When $n = 2$, the board is 4×4 and has four extra square symmetries. Exact enumeration of its 79 maximum placements gives fixed counts

$$(79, 3, 3, 7)$$

for the four operations specified in Section 2; their average is 23, the OEIS term. The two quarter-turns and two diagonal reflections have fixed counts

3, 3, 7, 7, so the full square group gives 14 orbits. Replacing only $a(2) = 23$ by 14 makes Barker's recurrence fail at $n = 10, 11, 12$. Therefore the fixed order-four convention in Section 2 is necessary, not cosmetic.

7. EXACT VERIFICATION AND DISCLOSURE

The compact serialized certificate records T , the rows z_i , all branch data, and the final rational function. Its verifier reconstructs the state set, transfer matrix, involutions, and the complete identity (7) from the board rules before comparing serialized data. A second verifier independently constructs a 68-dimensional integer linear representation of the four Burnside branches. If its transition matrix is B and input/output vectors are b, ℓ , it checks

$$\ell^T B^k D^{\text{rev}}(B)b = 0 \quad (0 \leq k < 68).$$

Cayley–Hamilton proves every later zero. That certificate also verifies an exact rational Bezout identity $UN + VD = 1$. A third, definition-level program enumerates the boundary cases and both group conventions.

All decisive calculations use integers or exact rational numbers and the Python standard library. OpenAI Codex assisted with proof discovery, code, adversarial checking, and manuscript preparation. Separate builder, counterexample-hunter, and referee roles were used, and the root verifier was implemented independently of the compact certificate. No proof assistant (Lean, Coq, Isabelle, or otherwise), external computer algebra system, floating-point inference, or randomized computation was used.

The complete source, certificates, verifiers, audit logs, and release manifest accompany this manuscript.

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